Eye Sentinel: training and data progress

Status as of November 11, 2026 | Glaucoma screening and image quality

Both disease models, the original quality model, and the EyeQ-trained quality model have completed training. Current work examines disease scores and image quality to design a review rule.

Data sources and roles

Stage	Sources and purpose	Images
Disease models	AIROGS-Light, ODIR-5K, Kaggle Eye Diseases	10,354 train; 1,190 validation
Original quality model	Disease images and FIQS; continuous scores from the FTHNet teacher	12,147 train; 1,642 validation
Boundary development	Disease images used to study p and q	1,230 after correction
Later tests	Internal test, ACRIMA, HYGD	1,233; 705; 737

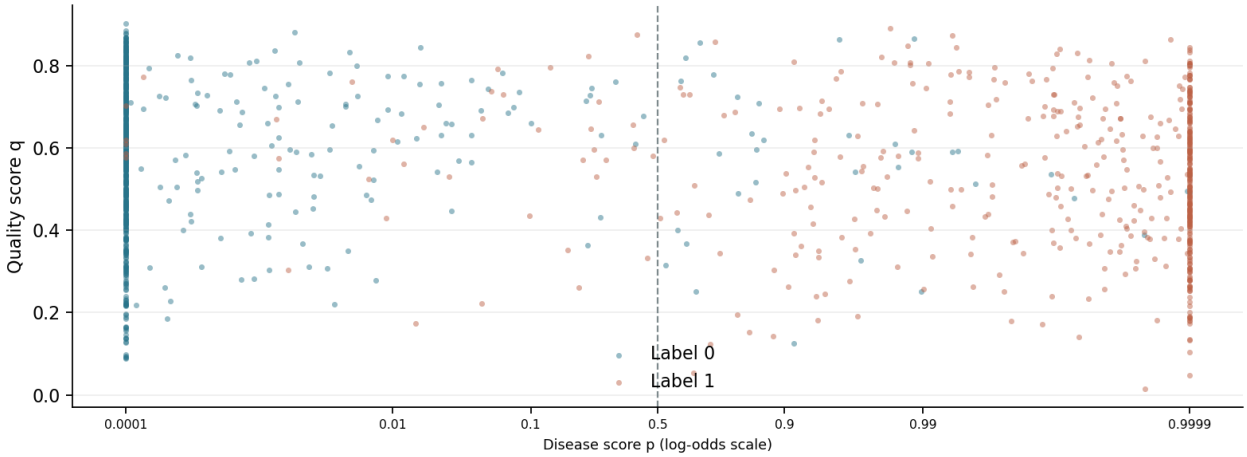
Disease label distribution

Data role	Label 0	Label 1	Total
Training	5,932	4,422	10,354
Model validation	701	489	1,190
Corrected boundary development	717	513	1,230
Internal test	744	489	1,233

Labels retain their source definitions. In AIROGS, 0 means no glaucoma referral and 1 means referral.

Initial disease and quality score distribution

Each point is a corrected development image. The horizontal axis is disease model A's score p, and the vertical axis is the original quality model's score q. The dashed line marks p = 0.5.



Development set by source

Source	Label 0	Label 1	Total
AIROGS-Light	393	390	783
ODIR-5K	277	28	305
Kaggle Eye Diseases	47	95	142

Two images associated with training or validation images were removed from the original 1,232-image boundary set. All three models scored the remaining 1,230 images.

Disease model training

Model A uses EfficientNet-B0 and model B uses ResNet50. Each trained for 20 epochs; checkpoints were selected using AIROGS model-validation AUROC.

Validation AUROC	AIROGS (769)	ODIR (277)	Kaggle (144)
A, epoch 16	0.9766	0.9042	0.9865
B, epoch 18	0.9831	0.8950	0.9782

Quality models and EyeQ

The original quality student learned continuous scores produced by a frozen FTHNet teacher for 13,789 images. Its selected epoch 16 had a teacher-fit Huber loss of 0.000517 on 1,642 validation images.

EyeQ supplies three human-assigned quality grades: Good, Usable, and Reject. The project acquired 28,792 original images, including 12,543 official training and 16,249 official test images. Image and known-patient association screening informed a separate EyeQ quality-training version.

EyeQ quality-model data	Train	Validation
All quality images	4,722	616
EyeQ images	1,007	117
Earlier quality data	3,715	499

The new EfficientNet-B0 quality model jointly learns FTHNet teacher scores and EyeQ grades. Training ran for 20 epochs, with epoch 17 selected by the combined validation loss.

Epoch 17 validation result	Value	Sample
EyeQ three-grade macro-F1	0.811	117 EyeQ images
EyeQ mean grade error	0.111	117 EyeQ images
Teacher-score MAE	0.0629	616 quality images
Combined validation loss	0.01009	616 quality images

Additional datasets and exploratory results

Additional images were examined for source, labels, and associations. The project downloaded and scored exploratory subsets from AIROGS, Chaksu, and PAPILA.

Source	Size and labels	Exploratory results
AIROGS originals	5,000: 4,837 no referral and 163 referral	682 images had no association candidate, all no referral; 113 had p_A >= 0.5
Chaksu	1,009: 872 normal and 137 suspect	265 images scored; 22 of 29 suspect images had p_A < 0.5
PAPILA	488 images from 244 people: 333 healthy, 87 glaucoma, 68 suspect	76 images scored; 10 of 13 glaucoma images had p_A < 0.5, from 7 people

The display threshold is 0.5. Chaksu suspect labels and PAPILA glaucoma diagnoses follow their respective source definitions.

Additional data acquired

Dataset	Contents	Current role
FIVES	800 images; 200 in each of four disease groups	Quality and disease analysis; 101 glaucoma images have at least one poor-quality attribute
MSHF	1,302 images, including 52 glaucoma images	Quality analysis; 50 glaucoma images have poor overall quality
RIM-ONE / BEH	485 / 634 images	Additional disease sources; duplicate and label review
Real Fundus / HRF	120 / 18 same-eye pairs of differing quality	Comparison of quality scores on real photographs

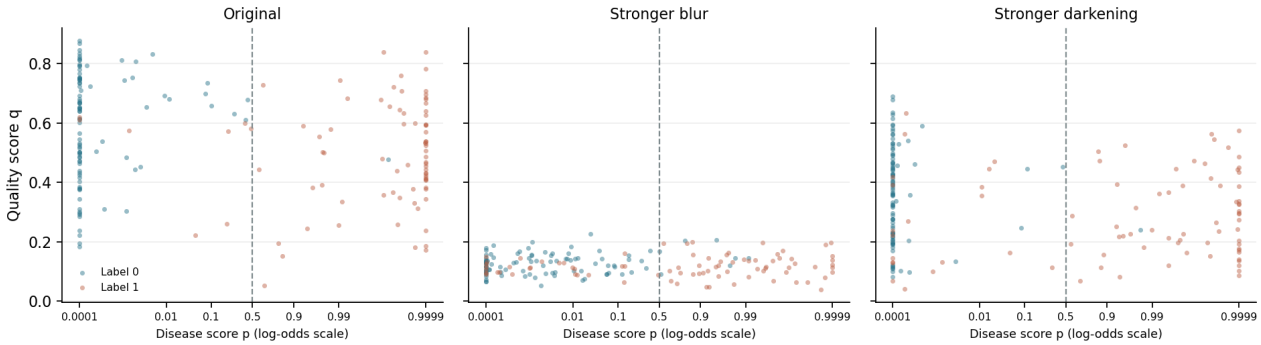
Degradation methods and paired results

To examine changes in image quality, 179 boundary-development images were selected: 83 with label 1 and 96 with label 0. Two levels each of Gaussian blur and darkening produced 716 versions. Disease model A and the original quality model scored each original and its variants.

Image version	Label 1 with p < 0.5	Label 0 with p >= 0.5	Median change in q
Original	8 / 83	1 / 96	0
Milder blur	14 / 83	12 / 96	-0.253
Stronger blur	27 / 83	7 / 96	-0.438
Milder darkening	12 / 83	2 / 96	-0.065
Stronger darkening	20 / 83	1 / 96	-0.214

Stronger blur moved 21 positive-label images from p >= 0.5 to below 0.5, including 16 below 0.1. Stronger darkening moved 13 below 0.5, including 12 below 0.1.

Score distribution for the same images



Blur radii were 2 and 5 pixels relative to a 512-pixel long edge. Brightness multipliers were 0.70 and 0.40.

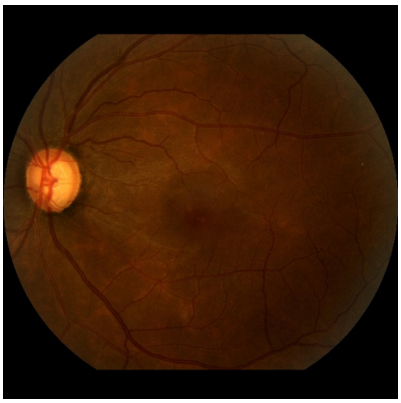
Real quality pairs and next steps

On real same-eye pairs, the original quality model assigned a higher q to the clearer image in the following numbers of pairs.

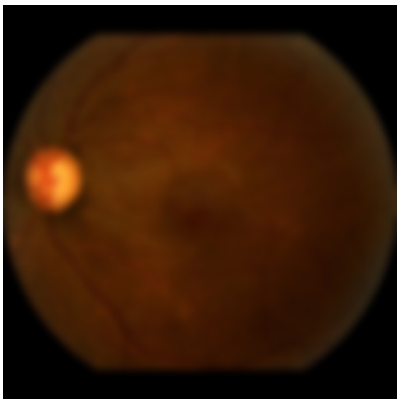
Dataset	Pairs	Clearer image scored higher	Median q difference (high - low)
Real Fundus	120	120 / 120	0.605
HRF	18	16 / 18	0.042

One image and its simulated versions

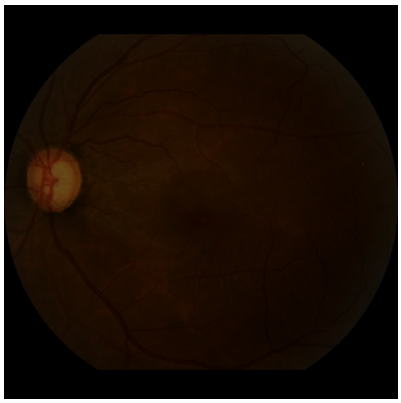
Original



Stronger blur



Stronger darkening



Next steps

1. Check disease and quality labels in FIVES and MSHF, including image associations and grading differences.

2. Use real same-eye photographs to set blur and exposure levels and assess optic-disc visibility.

3. Compare disease-score-only and quality-aware review at the same review volume, including missed positives and false positives.

4. Fix the review rule, then evaluate the internal test, ACRIMA, and HYGD sets.

Sources: formal_v3 training and checkpoint review, EyeQ training logs, corrected boundary scores, and data expansion and paired-image audits.